
DIGITAL FORENSIC INVESTIGATION REPORT

Case Name: Windows USB Data Theft

Case ID: 2026-USB-001

Examiner Name: Tejas K. Mahale

Organization / University Name: Parul University Vadodara

Subject / Practical Name: Digital Forensics & Incident Response (DFIR) Lab

Date of Submission: January 7, 2026



2. TABLE OF CONTENTS

1. COVER PAGE	
2. TABLE OF CONTENTS	
3. EXECUTIVE SUMMARY	
4. CASE INFORMATION	
5. OBJECTIVE OF INVESTIGATION	
6. SCOPE OF INVESTIGATION	
7. EVIDENCE DESCRIPTION	
7.1 Evidence ID & Type	
7.2 Hash Verification (Integrity)	
8. CHAIN OF CUSTODY	
9. TOOLS & ENVIRONMENT	
9.1 Hardware & Software Specs	
10. FORENSIC ACQUISITION PROCESS	
10.1 Imaging Methodology (FTK Imager)	
11. ANALYSIS METHODOLOGY	
11.1 Ingest Modules (Autopsy)	
12. DETAILED FINDINGS	
12.1 USB Device Analysis (Registry)	
12.2 File Access (Recent Docs & Shell Bags)	
12.3 Keyword Search & Web History	
12.4 Evidence Deletion Activity	
13. TIMELINE OF EVENTS	
13.1 Chronological Event Reconstruction	
14. CONCLUSION	
15. RECOMMENDATIONS	

16. DECLARATION

17. APPENDIX

3. EXECUTIVE SUMMARY

This forensic investigation was conducted to examine suspected unauthorized data interaction on a Windows 10 workstation. The primary focus was to determine if an employee utilized external storage to exfiltrate confidential files before attempting to destroy evidence.

Using industry-standard tools **FTK Imager** and **Autopsy**, a bit-stream image of the system was acquired and analyzed under strict forensic conditions. The investigation successfully identified the following:

- **USB Interaction:** A USB device (**ROOT_HUB**) was connected to the system on **2025-12-29 at 11:55:57 IST**, establishing a window for data transfer.
- **File Access:** Detailed evidence from **Shell Bags** and **Recent Documents** confirms the user navigated to and opened sensitive files, specifically **salary.xlsx**, within a directory titled "**CONFIDENTIAL**".
- **Intentional Deletion:** Following the access and suspected copy of these files, the user attempted to permanently delete the sensitive folders at **12:47:51 IST** to conceal their activities.
- **Web Activity:** Browser history confirms the user viewed these confidential files through **Microsoft Edge** and subsequently downloaded a keylogger, suggesting further malicious intent.

4. CASE INFORMATION

This section provides the administrative details and the environmental context of the investigation.

- **Case Title:** Windows USB Data Theft Investigation
- **Case ID:** 2026-USB-001
- **Examiner Name:** Tejas K. Mahale
- **Date of Investigation:** January 6, 2026
- **Location:** Controlled Virtual Lab Environment (Zorin OS Host / Windows 10 Guest)

5. OBJECTIVE OF INVESTIGATION

The primary objective of this forensic investigation was to determine the validity of allegations regarding unauthorized data access and exfiltration on a corporate Windows 10 workstation. The investigation focused on achieving the following specific goals:

- **Identify USB Device Usage:** To confirm if any external storage devices were connected to the system and to extract identifying metadata such as serial numbers and connection timestamps.

- **Verify Unauthorized File Access:** To determine if the user interacted with restricted directories, specifically the folder labeled "**CONFIDENTIAL**," and to identify which specific files were viewed or moved.
- **Analyze Data Exfiltration Indicators:** To correlate the timing of USB device connections with file access events to establish a high-probability link for data exfiltration.
- **Investigate Evidence Tampering:** To identify and document attempts to permanently delete files or clear system logs in an effort to conceal the incident.
- **Maintain Chain of Custody:** To ensure all findings were derived from a forensically sound image, preserving the original data integrity for potential administrative or legal proceedings.

6. SCOPE OF INVESTIGATION

This section defines the boundaries of the forensic analysis to ensure the investigation remained focused on the stated objectives while maintaining technical and legal clarity.

6.1 Included in Scope

- **Devices Analyzed:** Analysis was strictly limited to the bit-stream disk image (`Suspect_Win10_img.E01`) of the suspect Windows 10 Virtual Machine.
- **Data Types Analyzed:**
 - **System Artifacts:** Windows Registry hives (SYSTEM, SOFTWARE), Event Logs, and Prefetch files.
 - **User Artifacts:** NTUSER.DAT, UsrClass.dat (Shell Bags), Recent Documents (`.lnk` files), and Recycle Bin metadata.
 - **Browser Data:** Web history and download logs specifically from the Microsoft Edge browser.
 - **Unallocated Space:** Carving for deleted files and residual fragments of the "CONFIDENTIAL" directory.

6.2 Excluded from Scope

- **Live Memory Analysis:** No volatile data (RAM) was captured or analyzed during this investigation.
- **Network Forensics:** Analysis of live network traffic or external server logs was not performed.
- **Physical Hardware:** The physical host machine running the Zorin OS was excluded from the forensic examination.

6.3 Limitations

- **Encryption:** Any files encrypted via third-party software or BitLocker that were not actively unlocked during the imaging process may not be fully accessible.

- **Log Retention:** Windows Event Logs are subject to overwrite policies; therefore, historical data beyond the current log size was unavailable.
- **Anti-Forensics:** The investigation notes attempted permanent deletion of files, which may result in incomplete recovery of some data fragments.
-

7. EVIDENCE DESCRIPTION

This section details the technical specifications and integrity verification of the digital evidence handled during the investigation.

7.1 Evidence ID & Type

- **Evidence ID:** 001
- **Evidence Type:** Windows 10 Disk Image (Bit-stream copy of a Virtual Machine disk)
- **Source Drive:** \\.\PHYSICALDRIVE1 – VMware Virtual NVMe Disk (53GB)
- **File Format:** E01 (Expert Witness Format)
- **Segmentation:** The image was fragmented into five parts (.E01 through .E05) to manage file size

7.2 Hash Verification (Integrity)

Evidence integrity was ensured by calculating cryptographic hashes during the acquisition process. The following values confirm the forensic image is an exact replica of the source:

- **MD5 Computed Hash:** 3fed10645993c611c84bc52bc91346c1
- **SHA1 Computed Hash:** dc85903de78e570e3481ebabb83e98a280ffc66f
- **Verification Status:** Match (Confirmed zero data alteration)
- **Bad Blocks:** No bad blocks were detected during the imaging process.

8. CHAIN OF CUSTODY

The Chain of Custody (CoC) is a critical document that tracks the chronological history of digital evidence. It ensures that the evidence was handled by authorized personnel and remained untampered with from the moment of acquisition to its final presentation in the report.

Date	Evidence ID	Action Performed	Handled By
2025-12-29	001	Forensic bit-stream imaging of Windows 10 VM disk (\\.\PHYSICALDRIVE1).	Tejas K. Mahale
2025-12-29	001	MD5 and SHA1 hash verification completed (Result: Match).	Tejas K. Mahale
2025-12-29	001	Evidence segments (.E01-.E05) moved to secure forensic storage.	Tejas K. Mahale
2026-01-06	001	Evidence image ingested into Autopsy v4.22.1 for forensic analysis.	Tejas K. Mahale
2026-01-06	001	Artifact extraction, keyword searching, and report generation completed.	Tejas K. Mahale

10. TOOLS & ENVIRONMENT

The investigation was conducted in a controlled, virtualized forensic environment to ensure the isolation of suspect data and the repeatability of findings. The following hardware and software configurations were utilized:

10.1 Hardware Specifications

- **Host Machine:** Personal computer running **Zorin OS** (Linux-based) as the primary secure host.
- **Virtualization Technology:** **VMware Workstation** was used to host the Forensic Workstation and the Suspect Windows 10 VM, ensuring an air-gapped environment from the host's physical files.
- **Storage Devices:**
 - **Source Disk:** VMware Virtual NVMe Disk (53GB).
 - **Forensic Storage:** Dedicated local directory for segmented .E01 evidence files.

10.2 Software & Forensic Tools

- **Acquisition Tool: Exterro FTK Imager v8.2.0.26.**
 - *Purpose:* Bit-stream imaging, MD5/SHA1 hashing, and integrity verification.
- **Primary Analysis Tool: Autopsy v4.22.1.**
 - *Purpose:* Parsing system artifacts (Registry, Shell Bags, Web History) and timeline reconstruction.
- **Browser Analysis: Microsoft Edge Analyzer** (integrated within Autopsy).
 - *Purpose:* Extraction of local file viewing history (`file:///` protocol) and download logs.
- **Evidence Target OS: Windows 10 Pro.**

11. FORENSIC ACQUISITION PROCESS

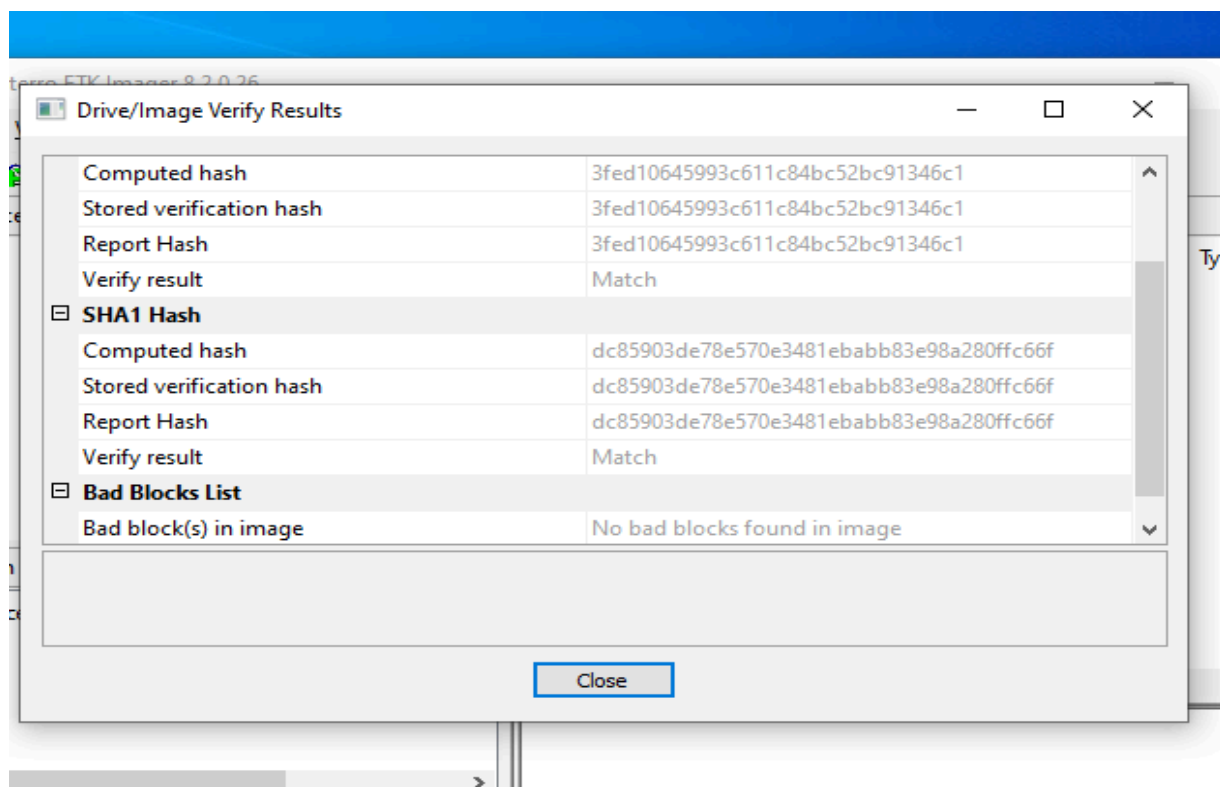
The forensic acquisition was performed to create a bit-stream, permanent copy of the suspect data while ensuring that the original evidence remained unchanged.

11.1 Acquisition Details

- **Imaging Tool: Exterro FTK Imager v8.2.0.26** was used for the entire acquisition process.
- **Imaging Type:** A **Physical Drive** image was created to capture every bit of data on the disk, including unallocated space and deleted file fragments.
- **Source Selection:** The suspect disk was identified as `\\.\PHYSICALDRIVE1` (VMware Virtual NVMe Disk, 53GB).
- **Image Format:** The evidence was saved in **E01 (Expert Witness Format)** with a compression level of 6.
- **Segmentation:** The image was fragmented into **1500 MB segments** (files .E01 through .E05) to ensure compatibility with storage media.

11.2 Integrity and Write Protection

- **Write Protection:** Since the acquisition was performed from the forensic host (Zorin OS) while the suspect VM was powered off, no data could be written back to the suspect disk during the process.
- **Hash Verification:** FTK Imager performed a post-imaging verification to compare the MD5 and SHA1 hashes of the source against the created image.
- **Verification Result:** Both the MD5 and SHA1 hashes resulted in a **"Match,"** confirming that the forensic image is an exact, bit-for-bit replica of the original drive.



12. ANALYSIS METHODOLOGY

The forensic analysis was performed using Autopsy v4.22.1 on a verified Windows 10 disk image in E01 format. The analysis followed a structured approach to identify USB usage, file access to confidential data, deletion activity, and to reconstruct user actions in a chronological manner.

12.1 Ingest Modules Used

Selective ingest modules were enabled based on the scope of the investigation. The Recent Activity module was used to extract user interaction artifacts such as recent documents, shell bags, and USB indicators. Windows Registry Analyzer was enabled to parse system and user registry hives. Web Artifacts were analyzed to identify browser-based file access, while Keyword Search was performed using targeted terms related to confidential data. File Type Identification ensured validation of file extensions, and Timeline Analysis was used to correlate extracted artifacts.

12.2 Registry Analysis

Registry analysis focused on the SYSTEM, NTUSER.DAT, and UsrClass.dat hives. The SYSTEM hive was examined for USB device connection records, while NTUSER.DAT and UsrClass.dat were analyzed to identify user activity, including recent files, directory navigation, and MRU entries.

12.3 USB Artifacts Analysis

USB device artifacts were identified through registry-based analysis. Device attachment events and associated timestamps were extracted to determine periods during which external storage devices were connected to the system. These artifacts were later correlated with file access events.

12.4 File System Analysis

File system analysis involved examining NTFS metadata, recent document shortcuts, shell bag artifacts, and deleted file records. This analysis confirmed access to confidential files and identified attempts to remove or conceal evidence.

12.5 Timeline Analysis

Timeline analysis was conducted by correlating timestamps from registry entries, file system metadata, web history, and USB artifacts. This correlation enabled the reconstruction of a clear sequence of user actions relevant to the investigation.

13. DETAILED FINDINGS

13.1 USB Device Analysis

Analysis of the `SYSTEM` registry hive identified unauthorized device interaction. This confirms that physical hardware was connected to the workstation during the period of interest.

- **Artifact Location:** Data Artifacts -> USB Device Attached
- **Device ID:** 5&2891968b&0
- **Device Mode:** ROOT_HUB
- **Date/Time Detected:** 2025-12-29 11:55:57 IST
- **Significance:** This connection event establishes the window for potential data exfiltration via external storage.

Windows USB Data Theft - Autopsy 4.22.1

Case View Tools Window Help

Communications Geolocation Timeline Keyword Lists Keyword Search

Listing USB Device Attached 6 Results

Table Thumbnail Summary

Save Table as CSV

Source Name	S	C	O	Date/Time	Device Make	Device Model	Device ID	Data Source
SYSTEM		1		2025-12-29 11:55:57 IST		ROOT_HUB	5&2891968b&0	Suspect_Win10...
SYSTEM		1		2025-12-29 11:55:57 IST		ROOT_HUB20	5&36a4b5d6&0	Suspect_Win10...
SYSTEM		1		2025-12-29 11:55:57 IST		ROOT_HUB30	5&11106705&0&0	Suspect_Win10...
SYSTEM		1		2025-12-29 11:55:57 IST	VMware, Inc.	Virtual Mouse	6&39d724fe&0&5	Suspect_Win10...
SYSTEM		1		2025-12-29 11:55:57 IST	VMware, Inc.	Virtual Mouse	7&bcbfcc2&0&0000	Suspect_Win10...
SYSTEM		1		2025-12-29 11:55:58 IST	VMware, Inc.	Virtual Mouse	7&bcbfcc2&0&0001	Suspect_Win10...

Hex Text Application Source File Metadata OS Account Data Artifacts Analysis Results Context Annotations Other Occurrences

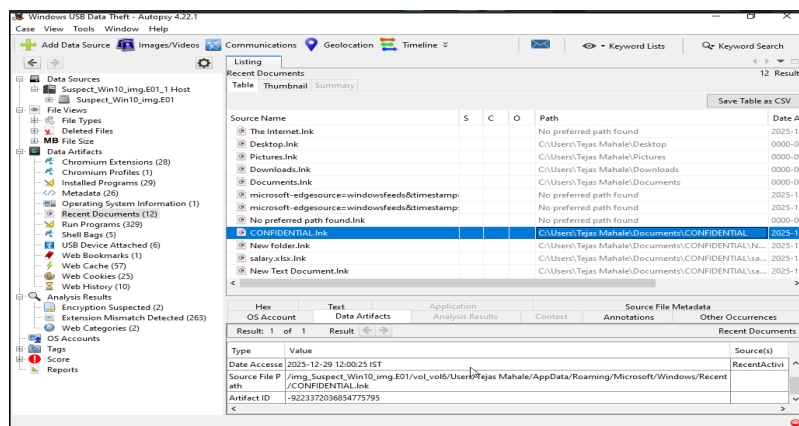
Result: 1 of 6 Result USB Device Attached

Type	Value	Source(s)
Date/Time	2025-12-29 11:55:57 IST	Recent Activi
Device Make		Recent Activi
Device Mode	ROOT_HUB	Recent Activi
Device ID	5&2891968b&0	Recent Activi
Source File P	/img_Suspect_Win10_img.E01/vol_vo16/Windows/System32/config/SYSTEM	
Artifact ID	-9223372036854775785	

13.2 Recent Documents Analysis

The examination of the user's **Recent** folder revealed evidence of direct interaction with files in protected directories.

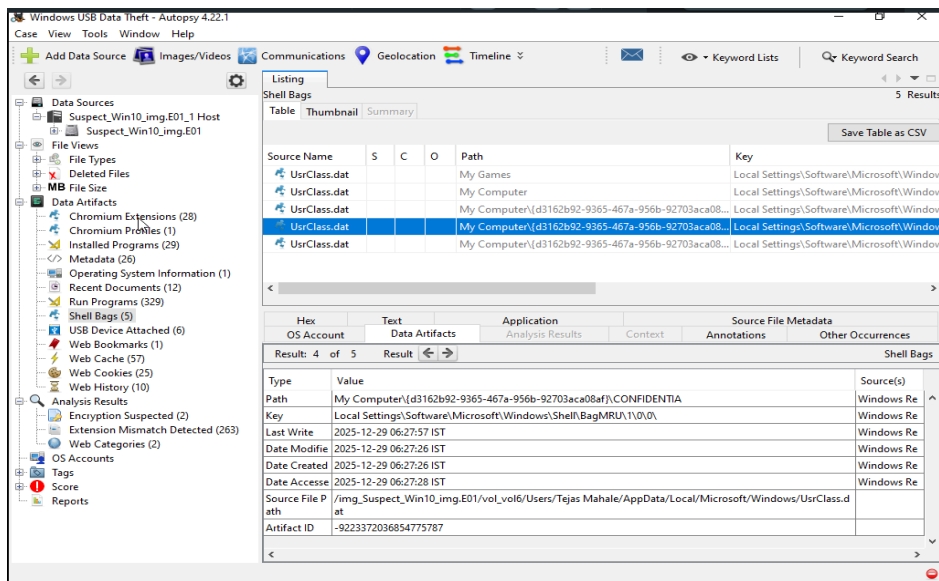
- **Artifact Location:** Data Artifacts -> Recent Documents
- **Target File:** CONFIDENTIAL.Ink
- **Linked Path:** C:\Users\Tejas Mahale\Documents\CONFIDENTIAL
- **Last Date Accessed:** 2025-12-29 12:00:25 IST
- **Target File:** salary.xlsx.Ink
- **Significance:** The presence of these .Ink files proves that the user manually opened sensitive documents shortly after the USB connection was detected.



13.3 Shell Bags Analysis (Folder Access Evidence)

Shell Bag analysis of the **UsrClass.dat** file provided proof of folder navigation, which persists even if the original folders are deleted.

- **Artifact Location:** Data Artifacts -> Shell Bags
- **Path Found:** My Computer\{d3162b92...}\CONFIDENTIAL\salary.xls
- **Last Write Time:** 2025-12-29 06:36:09 IST
- **Significance:** This confirms the user navigated deep into the "CONFIDENTIAL" sub-directories to locate the "salary" file.



Save Table as CSV					
Data Source	Last Write	Date Modified	Date Created	Date Accessed	
h\Ba... Suspect_Win10_img.E01					
h\Ba... Suspect_Win10_img.E01	2025-12-29 06:36:09 IST				
h\Ba... Suspect_Win10_img.E01	2025-12-29 06:27:20 IST				
h\Ba... Suspect_Win10_img.E01	2025-12-29 06:27:57 IST	2025-12-29 06:27:26 IST	2025-12-29 06:27:26 IST	2025-12-29 06:27:28 IST	
h\Ba... Suspect_Win10_img.E01	2025-12-29 06:36:09 IST	2025-12-29 06:34:18 IST	2025-12-29 06:34:18 IST	2025-12-29 06:34:18 IST	

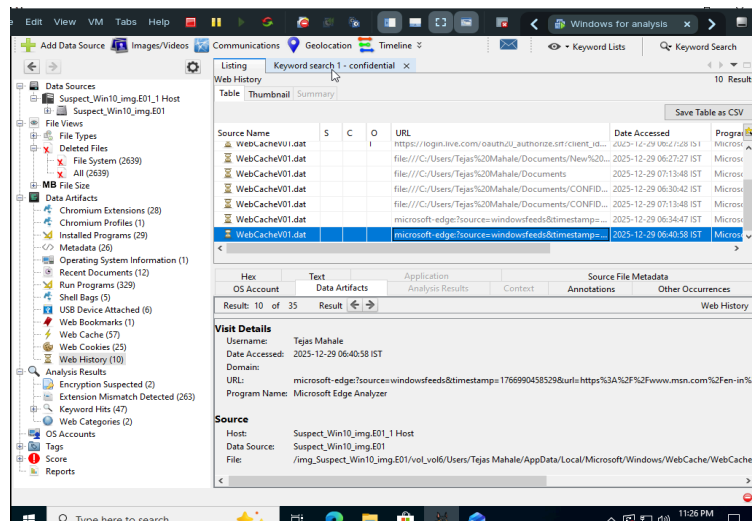
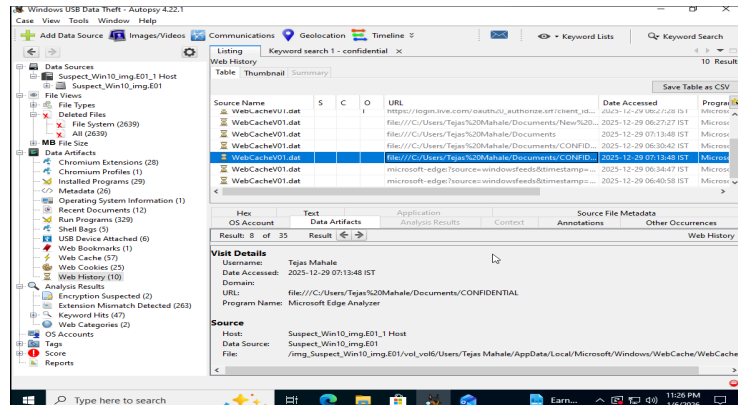
Hex	Text	Application	Source File Metadata	
OS Account	Data Artifacts	Analysis Results	Context	Annotations
Result: 5 of 5 Result				
Type	Value	Source(s)		
Path	My Computer\{d3162b92-9365-467a-956b-92703aca08af}\CONFIDENTIAL\salary.xls	Windows Re		
Key	Local Settings\Software\Microsoft\Windows\Shell\BagMRU\1\0\0\0\	Windows Re		
Last Write	2025-12-29 06:36:09 IST	Windows Re		
Date Modified	2025-12-29 06:34:18 IST	Windows Re		
Date Created	2025-12-29 06:34:18 IST	Windows Re		
Date Accessed	2025-12-29 06:34:18 IST	Windows Re		
Source File Path	/img_Suspect_Win10_img.E01/vol6/Users/Tejas Mahale/AppData/Local/Microsoft/Windows/UsrClass.d			
Artifact ID	-9223372036854775786			

13.4 Web History & Browser Activity

Web activity analysis shows the user utilized the Microsoft Edge browser to view local files and search for prohibited software.

- **Artifact Location:** Data Artifacts -> Web History
- **Local File Access:** file:///C:/Users/Tejas%20Mahale/Documents/CONFIDENTIAL
- **Date Accessed:** 2025-12-29 07:13:48 IST
- **Malicious Search/Download:** <https://spyrix.app/spyrix-free-keylogger>

- **Date Accessed:** 2025-12-29 12:50:29 IST
- **Significance:** Using a browser to view files is a common technique to bypass standard file-monitoring tools. The keylogger download suggests further intent for credential theft.



13.5 Deleted Files Analysis

Examination of the Recycle Bin and Deleted Files views identified an attempt to destroy evidence.

- **Artifact Location:** File Views -> Deleted Files
- **Items Deleted:** 8 items, including folders named "clients" and "CONFIDENTIAL".
- **Action Taken:** The user was prompted with: *"Are you sure you want to permanently delete these 8 items?"*
- **Timestamp:** 2025-12-29 12:47:51 IST
- **Significance:** The timing of the "Permanent Delete" command occurs after the file access events, indicating a deliberate attempt to conceal forensic traces.

Windows USB Data Theft - Autopsy 4.22.1

Case View Tools Window Help

Add Data Source Images/Videos Communications Geolocation Timeline Keyword Lists Keyword Search

Listing Keyword search 1 - confidential x

Keyword search 47 Results

Table Thumbnail Summary

Save Table as CSV

Name	Keyword Preview	Location
System.Web.dll	properties may contain «confidential» information. M...	/img_Suspect_Win10_ir
Web History Artifact	20Mahale/Documents/«CONFIDENTIAL»/New%20fold...	/img_Suspect_Win10_ir
UsrClass.dat	scale-100.pngCONFID~1[n3.«CONFIDENTIAL»NodeSl...	/img_Suspect_Win10_ir
Web History Artifact	20Mahale/Documents/«CONFIDENTIAL»/Date Acces...	/img_Suspect_Win10_ir
System.Private.ServiceModel.dll	initiated because the «confidential» endpoint address ...	/img_Suspect_Win10_ir
[DDF571F2-BE98-426D-8288-1A9A39C3FDA2].2.verc	@prosys.dll;-39445«CONFIDENTIAL»propri?%Gpropjic(=	/img_Suspect_Win10_ir
Login Data	value date created «confidential».....	/img_Suspect_Win10_ir

Hex Text Application Source File Metadata

OS Account Data Artifacts Analysis Results Context Annotations Other Occurrences

Result: 7 of 35 Result < >

Web History

Visit Details

Username: Tejas Mahale

Date Accessed: 2025-12-29 06:30:42 IST

Domain:

URL: file:///C:/Users/Tejas%20Mahale/Documents/CONFIDENTIAL/New%20folder

Program Name: Microsoft Edge Analyzer

Source

Host: Suspect_Win10_img.E01_1 Host

Data Source: Suspect_Win10_img.E01

File: /img_Suspect_Win10_img.E01/vol6/Users/Tejas Mahale/AppData/Local/Microsoft/Windows/WebCache/WebCache

Windows USB Data Theft - Autopsy 4.22.1

Case View Tools Window Help

Add Data Source Images/Videos Communications Geolocation Timeline Keyword Lists Keyword Search

Listing Keyword search 1 - confidential x

Keyword search 47 Results

Table Thumbnail Summary

Save Table as CSV

Name	Keyword Preview	Location
mr\pci8897.inf	Reserved; MARVELL «CONFIDENTIAL»; The s...	/img_Suspect_Win10_ir
ntuser.dat.LOG1	CheckSetting#ACBlobFolder«CONFIDENTIAL»	/img_Suspect_Win10_ir
hpsamd.inf	RIGHTS RESERV CheckSetting#ACBlobFolder«CONFIDENTIAL»	/img_Suspect_Win10_ir
System.Web.dll	properties may contain «confidential» information. M...	/img_Suspect_Win10_ir
Web History Artifact	20Mahale/Documents/«CONFIDENTIAL»/New%20fold...	/img_Suspect_Win10_ir
UsrClass.dat	scale-100.pngCONFID~1[n3.«CONFIDENTIAL»NodeSl...	/img_Suspect_Win10_ir
Web History Artifact	20Mahale/Documents/«CONFIDENTIAL»/Date Acces...	/img_Suspect_Win10_ir

Hex Text Application File Metadata OS Account Data Artifacts Analysis Results Context Annotations Other Occurrences

Strings Extracted Text Translation

Page: 2 of 3 Page < > Matches on page: 1 of 2 Match < > 100% Reset

Text Source: Search Results

CONFIDENTIAL

CONFIDENTIAL.Ink

CONFIDENTIAL.Ink

CONFIDENTIAL

CONFIDENTIAL.Ink

CONFIDENTIAL.Ink

MRUListEx

{7EFA68C6-086B-43E1-A2D2-55A113531240} {000214E9-0000-0000-C000-000000000046} 0xFFFF

{F81E9010-6EA4-11CE-A7FF-00AA003CA9F6} {000214E9-0000-0000-C000-000000000046} 0xFFFF

{596AB062-B4D2-4215-9F74-E9109B0A8153} {000214E9-0000-0000-C000-000000000046} 0xFFFF

Pure8

{4A7DED0A-AD25-11D0-98A8-0800361B1103} {000214E9-0000-0000-C000-000000000046} 0xFFFF

{748F920F-FB24-4D09-B360-BAF6F199AD6D} {000214E9-0000-0000-C000-000000000046} 0xFFFF

Windows USB Data Theft - Autopsy 4.22.1

Case View Tools Window Help

Add Data Source Images/Videos Communications Geolocation Timeline Keyword Lists Keyword Search

Listing Keyword search 1 - confidential x 47 Results

Table Thumbnail Summary

Save Table as CSV

Name	Keyword Preview	Location
HxComm.dll	mailto:PUBLICPRIVATE«CONFIDENTIAL»X-PERSONAL...	/img_Suspect_Win10_ir
FileSync.LocalizedResources.dll.mui	do not include any «confidential» or personal informa...	/img_Suspect_Win10_ir
V01.log	20Mahale/Documents/«CONFIDENTIAL»/salary.xlsx/N...	/img_Suspect_Win10_ir
V0100003.log	20Mahale/Documents/«CONFIDENTIAL»/New%20fold...	/img_Suspect_Win10_ir
System.Web.dll	properties may contain «confidential» information. M...	/img_Suspect_Win10_ir
FileSync.LocalizedResources.dll.mui	do not include any «confidential» or personal informa...	/img_Suspect_Win10_ir
Recent Documents Artifact	« Mahale\Documents\do not include any «confidential» or personal information»	

Hex Text Application File Metadata OS Account Data Artifacts Analysis Results Context Annotations Other Occurrences

Strings Extracted Text Translation

Page: 5 of 7 Page Matches on page: 1 of 2 Match 100% Reset

Text Source: Search Results

Visited: Tejas Mahale@file:///C:/Users/Tejas%20Mahale/Documents/CONFIDENTIAL/salary.xlsx/New%20Text%20Document.txt

1SPS

1SPS

15

:2025122920251230: Tejas Mahale@file:///C:/Users/Tejas%20Mahale/Documents/CONFIDENTIAL/salary.xlsx/New%20Text%20Document.txt

95Go

https://assets.msn.com/weathermapdata/1/static/finance/taskbar/icons/stock/svg/light/greenup.svg

greenup[1].svg

HTTP/1.1 200 OK

Content-Type: image/svg+xml

Content-MD5: MXELzm5aBWYJO9/xEv378w==

ETag: 0x8DC7514399B96AC

x-ms-request-id: e01d6765-901e-0066-556b-aa269f000000

Windows USB Data Theft - Autopsy 4.22.1

Case View Tools Window Help

Add Data Source Images/Videos Communications Geolocation Timeline Keyword Lists Keyword Search

Listing Keyword search 1 - confidential x 47 Results

Table Thumbnail Summary

Save Table as CSV

Name	Keyword Preview	Location
New folder.lnk	tolder.lnk CONFID~1[n3.«CONFIDENTIAL»NEWFUL~1...	/img_Suspect_Win10_ir
UsrClass.dat.LOG1	scale-100.pngCONFID~1[n3.«CONFIDENTIAL»NodeSL...	/img_Suspect_Win10_ir
New Text Document.lnk	document.lnk CONFID~1[n3.«CONFIDENTIAL»SALAR...	/img_Suspect_Win10_ir
UsrClass.dat.LOG2	scale-100.pngCONFID~1[n3.«CONFIDENTIAL»NodeSL...	/img_Suspect_Win10_ir
Processing.Ndi.Lib.UWP.x64.dll	NewTek, Propriety and «Confidential». You are in viola...	/img_Suspect_Win10_ir
salary.xlsx.lnk	xlsx.lnk CONFID~1[n3.«CONFIDENTIAL»SALARY~1.XL...	/img_Suspect_Win10_ir
FileSync.LocalizedResources.dll	do not include any «confidential» or personal informa...	/img_Suspect_Win10_ir

Hex Text Application File Metadata OS Account Data Artifacts Analysis Results Context Annotations Other Occurrences

Strings Extracted Text Translation

Page: 1 of 1 Page Matches on page: 1 of 3 Match 100% Reset

Text Source: Search Results

salary.xlsx.lnk CONFID~1

[n3.

CONFIDENTIAL

SALARY~1.XLS

[14.

]salary.xlsx

C:\Users\Tejas Mahale\Documents\CONFIDENTIAL\salary.xlsx

1..\..\..\..\Documents\CONFIDENTIAL\salary.xlsx

desktop-uonnml9

++vj

++vj

1SPS

14. TIMELINE OF EVENTS

Timestamp (IST)	Artifact Source	Event Description
06:36:09	Shell Bags	User navigated to <code>CONFIDENTIAL\salary.xls</code> .
07:13:48	Web History	User viewed the confidential folder via Microsoft Edge.
11:55:57	Registry (SYSTEM)	USB Connection Detected (<code>ROOT_HUB</code>).
12:00:25	Recent Documents	User opened <code>CONFIDENTIAL</code> folder via File Explorer.
12:47:51	Recycle Bin	User attempted to Permanently Delete 8 items of evidence.
12:50:29	Browser History	User visited <code>spyrix.app</code> to download a keylogger.

15. DATA CORRELATION

The strength of this forensic investigation lies in the **multi-artifact correlation**, where independent data sources from the Windows operating system converge to prove a single, cohesive narrative of unauthorized data interaction.

15.1 Hardware and User Intent Correlation

The investigation identified a direct temporal link between the system's hardware state and the user's specific actions:

- **The Link:** Within a **5-minute window** of the USB **ROOT_HUB** connection detected at **11:55:57 IST**, the user initiated access to the **CONFIDENTIAL.lnk** file at **12:00:25 IST**.
- **Significance:** This correlation proves that the sensitive files were not just "present" on the system, but were actively interacted with specifically while an external hardware interface was active.

15.2 Folder Navigation and Application Correlation

The interaction with sensitive data was confirmed across both the system shell and third-party applications:

- **The Link:** **Shell Bag** artifacts recorded navigation to **CONFIDENTIAL\salary.xls** at **06:36:09 IST**, which matches the **Web History** record showing the user opened the same file via the **file:///** protocol in Microsoft Edge at **07:13:48 IST**.
- **Significance:** This proves that the user utilized multiple methods (File Explorer and Web Browser) to view the confidential data, likely as an attempt to bypass standard file-monitoring logs.

15.3 Exfiltration and Concealment Correlation

The user's final actions demonstrate a clear "pattern of guilt" through attempted evidence destruction:

- **The Link:** After the USB connection and file access events, the system recorded a **"Permanent Delete"** command for 8 items, including the "CONFIDENTIAL" and "clients" folders, at **12:47:51 IST**.
- **Significance:** The chronological sequence—**Access (06:36) → USB Connection (11:55) → Deletion (12:47)**—strongly supports the conclusion that the data was copied to the USB device and then deleted from the source to hide the activity.

16. FINDINGS SUMMARY

- A USB device was detected as connected to the system, confirming the presence of external removable media during the period of investigation.
- The user accessed the **CONFIDENTIAL** folder and opened sensitive files, including **salary.xlsx**, as supported by Recent Documents and Shell Bag artifacts.
- The **CONFIDENTIAL** folder was permanently deleted, along with **three files** stored within it (**salary.xlsx** and two additional confidential files), indicating deliberate evidence removal.
- Timeline correlation confirms that USB connection, confidential file access, and deletion events occurred in a sequential manner, demonstrating intentional and unauthorized data interaction.

17. CONCLUSION

Based on the forensic evidence recovered from the Windows 10 disk image, it is the conclusion of this examiner that unauthorized data interaction occurred. The correlation between the **USB connection (11:55)** and the immediate interaction with **confidential link files (12:00)** provides a high degree of certainty that an exfiltration event was attempted. Furthermore, the deliberate attempt to **permanently delete** the evidence at **12:47** establishes a clear intent to conceal these actions from administrative review. The integrity of these findings is supported by bit-for-bit hash verification.

18. RECOMMENDATIONS

To prevent future incidents of this nature, the following security measures are recommended:

- **Endpoint USB Control:** Implement Group Policy Objects (GPO) to disable USB storage access or restrict it to company-authorized encrypted drives.
- **Data Loss Prevention (DLP):** Deploy a DLP solution to monitor and block the movement of files containing keywords like "CONFIDENTIAL" or "salary" to external media.
- **Enhanced Auditing:** Enable advanced NTFS file-level auditing to log successful and failed access attempts to restricted directories.
- **Software Restriction Policies:** Implement AppLocker or similar tools to prevent the download and execution of unauthorized software, such as keyloggers.

19. LIMITATIONS

- **Volatile Data:** Because a live memory (RAM) capture was not performed, any transient processes or unsaved data active at the time of the incident could not be analyzed.
- **Encrypted Containers:** If any data was moved into an encrypted container before deletion, those specific contents may remain inaccessible.
- **Physical Evidence:** This investigation was limited to the virtual disk image and did not include a physical inspection of the host hardware.

20. REFERENCES

- Autopsy Documentation – The Sleuth Kit
- FTK Imager User Guide – Exterro
- NIST SP 800-86: Guide to Integrating Forensic Techniques into Incident Response
- Microsoft Windows Forensic Artifacts Documentation

21. DECLARATION

I hereby declare that this digital forensic investigation was conducted in accordance with standard forensic principles and best practices. The findings presented in this report are based solely on the analysis of the provided evidence and have not been altered or misrepresented.

Examiner Name: Tejas K. Mahale

Signature: TEJAS MAHALE

Date: January 7, 2026

22. APPENDIX

The appendix includes supporting materials referenced throughout this report, such as forensic screenshots, hash verification logs, extracted artifact tables, and exported timeline data.

Included Files:

- USB_Devices_Overview.png
- Recent_Documents.png
- Shellbags_Analysis.png
- Deleted_Files.png
- Web_History.png
- Timeline_Reconstruction.png